SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA510I

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 **Total Marks:50**

Annexure A

CONCEPT MAPPING

Code of Conduct:

I S Madhuri (192465036) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

S Madhuri
Signature of the student:



Annexure A

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:
    a. Show how each attack leads to specific threats and risks
    b. Map how security goals help mitigate them Provide one real-world example for each link

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
    a. Show differences and similarities
        i.Map how each affects the Data structure and Security level
        ii.Analyze which is more vulnerable to attacks and why

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
    a. Show relationships between these concepts
    b. Analyze how each contributes to secure communication
    c. Include one real-world application for each

5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
    a. Show how each concept is mathematically connected
    b. Map their role in encryption/decryption
    c. Analyze which concept is most critical for public key cryptography and justify

RUBRICS FOR CALCULATION:

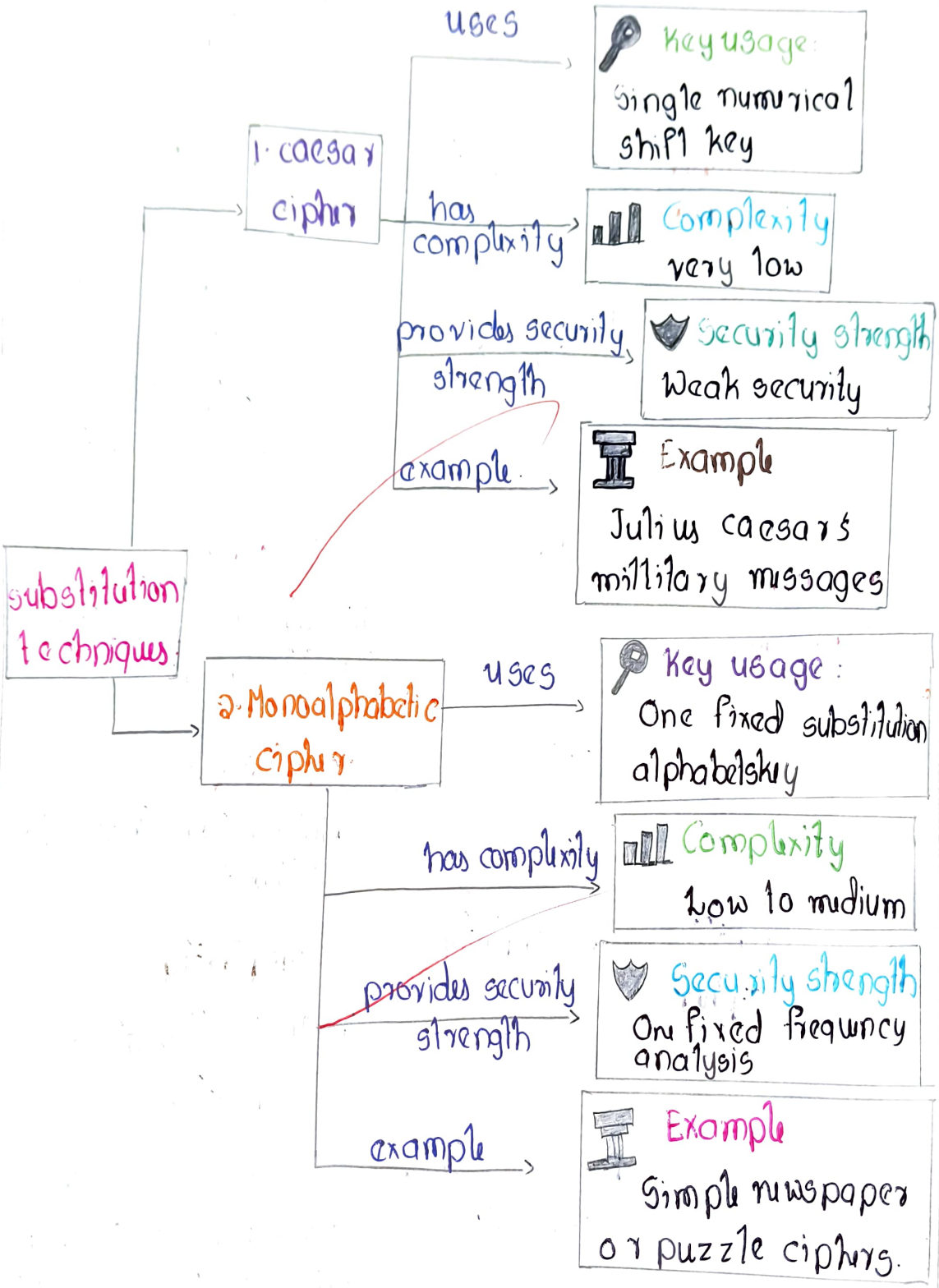
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25 23	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30 30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20 20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15 15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10 10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

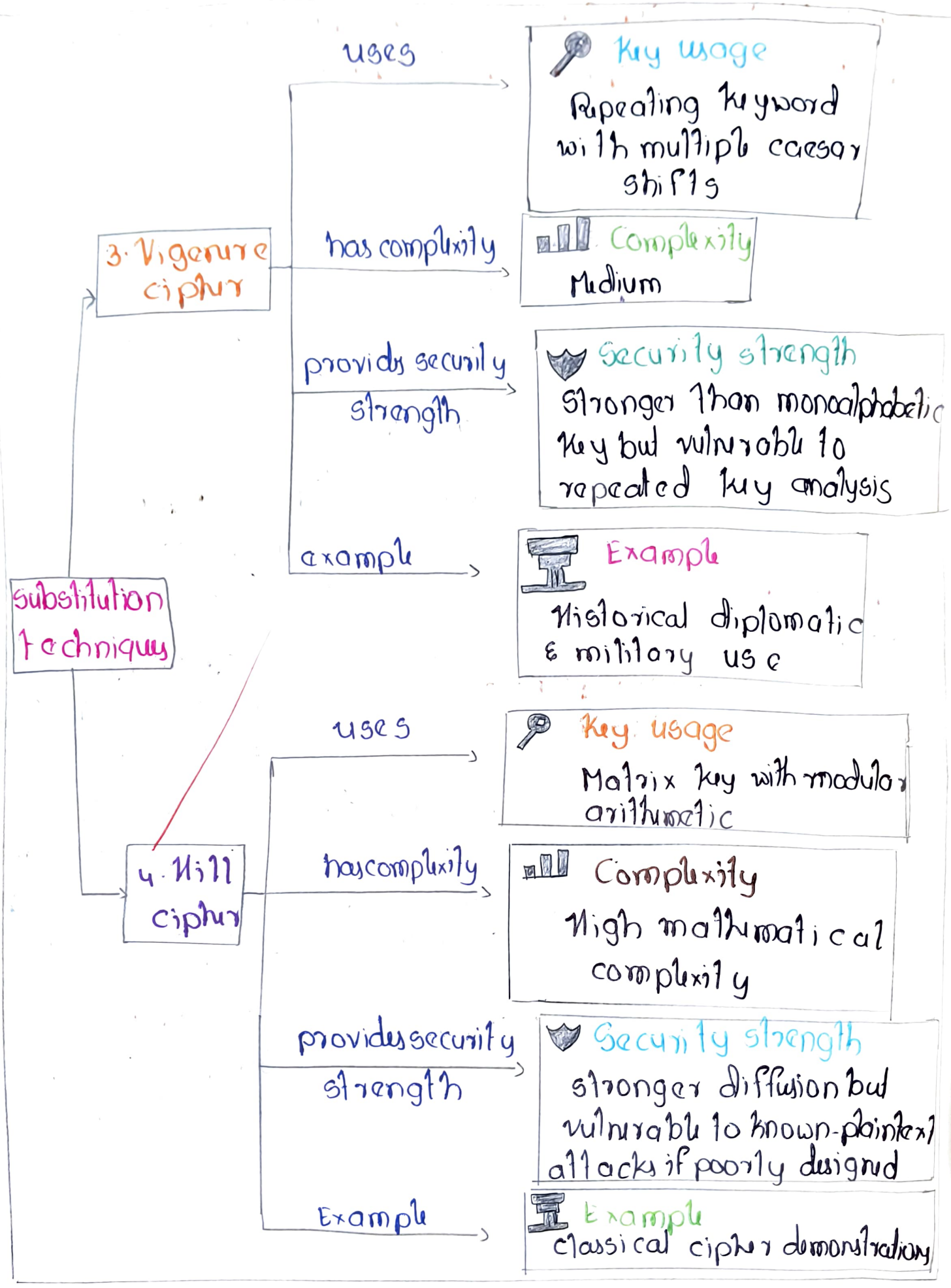
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98/100

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Substitution Techniques: Caesar Cipher, Monophabetic cipher, Vigenere cipher & hill cipher.




Substitution techniques
3. Vigenere cipher
uses
Key usage
Repeating keyword with multiple caesar shifts
has complexity
Complexity
Medium
provides security strength
Security strength
Stronger than monoalphabetic key but vulnerable to repeated key analysis
example
Example
Historical diplomatic & military use
4. Hill cipher
uses
Key usage
Matrix key with modular arithmetic
hascomplexity
Complexity
High mathematical complexity
providessecurity strength
Security strength
stronger diffusion but vulnerable to known-plaintext attacks if poorly designed
Example
Example
classical cipher demonstrations

Technique	Caesar cipher	Monoalphabetic substitution	Vigenere cipher	Hill cipher
Key usage	One shift key	One fixed substitution alphabet	Repeating keyword & multiple shifts	Invertible matrix key modulo m
complexity	very low	low-medium	medium	high
security strength	very weak	weak-medium	medium to strong	strong diffusion but not unconditionally secure
Main weakness	Brute force & frequency clues	frequency analysis	Repeated-key & frequency analysis	Diplomatic messages.
Example	Julius caesar	puzzle ciphers, newspaper cipher	Diplomatic messages	Classrom & classical cryptography demonstration

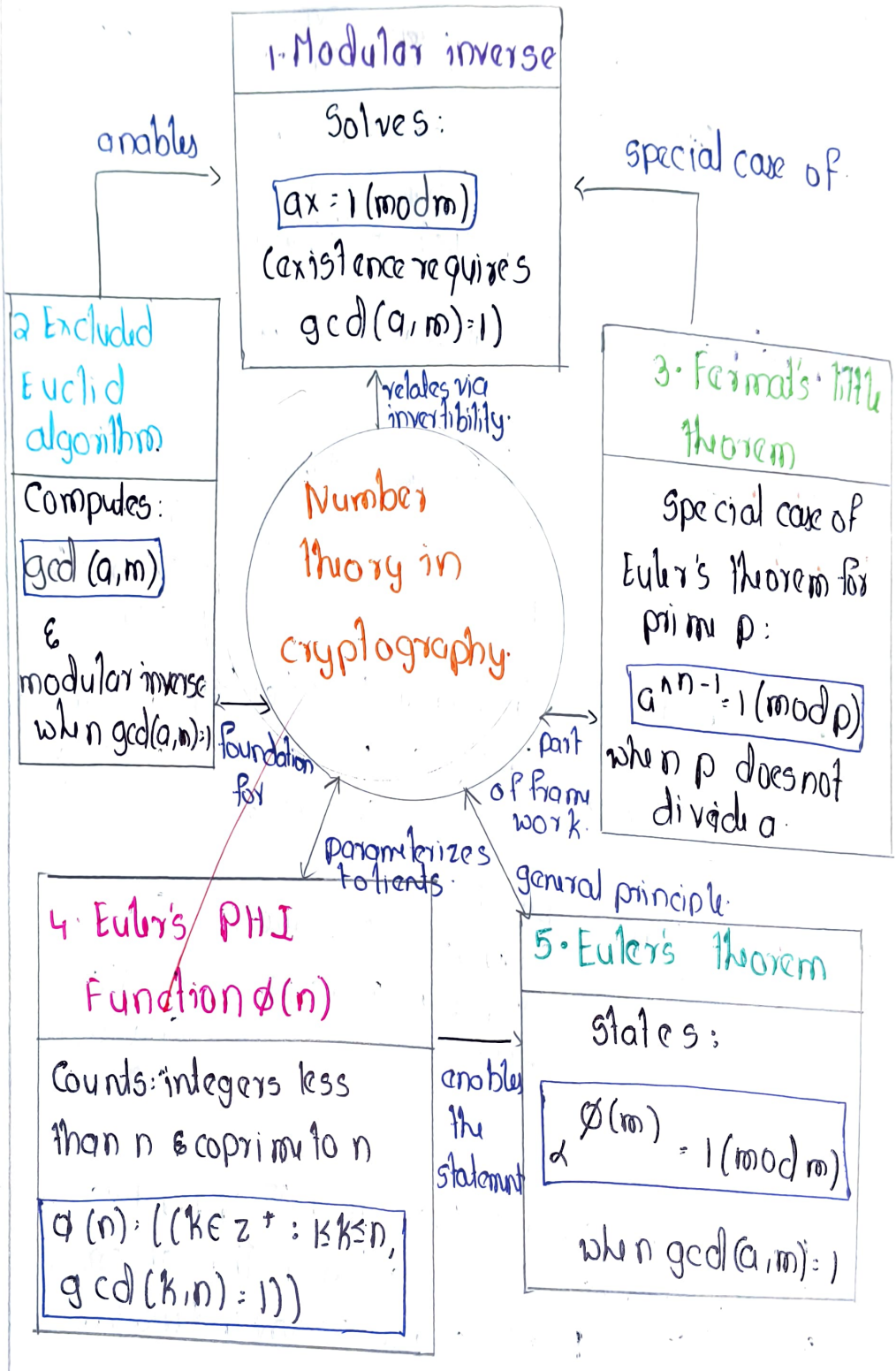
Conclusion:

Most secure among these is hill cipher because it encrypts blocks & provide diffusion through matrix multiplixation. But it is not secure in modern cryptoanalysis.

Relative security ranking: Worst->best

caesar cipher → monoalphabetic substitution → vigenere cipher → hill cipher.

5. Number theory concepts used in cryptography



cryptographic applications

Modular inverse

Used for

RSA private exponent & decryption

$m = a^d \pmod n$

Euler's PHI Function $\phi(N)$

Used for

RSA key generation

$n = pq,$

$\phi(n) = (p-1)(q-1)$

Euler's theorem

Used for

prove RSA correctness

$m^{r\phi(n)} = 1 \pmod n$

For $\gcd(n, n) = 1$

<1>

Extended euclid algorithm

Used to

find private exponent d

$ed = 1 \pmod{\phi(n)}$

a^k

Fermat's little algorithm

Used to

support modular exponentiation & prime field reasoning

$a^{p-1} = 1 \pmod p$

Mathematical chain:

<1> Extended Euclid algorithm ⟶ Modular inverse

ϕ Euler's Phi function ⟶ Euler's theorem

Fermat's little theorem ⟶ prime number special case of Euler's theorem

Encryption / Decryption role

- $\phi(n)$ helps generate RSA keys
- Extended euclid computes d as the modular inverse of e modulo $\phi(n)$
- Modular inverse enables recovery of the private exponent
- Euler's theorem supports the proof that RSA decryption reverses encryption
- Fp Fermat's theorem supports calculations in prime fields & explains public key modular exponentiation

Criticality analysis:

conclusion: Modular inverse

→ It is the operational step that produces the private exponent in RSA & other public-key systems

→ Without a valid modular inverse, encryption & decryption keys cannot be correctly paired.

Qualification:

Euler's phi function & euler's theorem are foundational for RSA mathmatical correctness.

Example

If $e \cdot d = 1 \bmod \phi(n)$, then d is the inverse of e modulo $\phi(n)$.

Conclusion:

Euler's phi function & Euler's Theorem

Basics for the relationships b/w exponents in modular arithmaticcs

RSA correctness

Use of Euler's theorem & $\phi(n)$ ensures that decryption recovers the original message.

Extended euclid

Computes gcd & coefficients that express the relationship b/w integers.

Modular inverse

$a^{-1} \bmod n$

Exists when $\gcd(a, n) = 1$; compute d using extended euclidean algorithm.

Private exponent

Modular inverse is used to find the private exponent d such that $ed = 1 \pmod{\phi(n)}$

Conclusion

Modular inverse is the most critical operational concept for public-key cryptography, supported by Euler's Theorem & phi function.